Raven Rothkopf

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Research Interests: AI-Assisted Programming, End-User Programming, Neurosymbolic Program Synthesis

Education

University of California, San Diego — La Jolla, CA
PhD in Computer Science; GPA: 3.9 — *Fall 2024 – Present*

Barnard College, Columbia University — New York, NY
BA in Computer Science; GPA: 3.8 — *Graduated Spring 2024*

Research Experience

Graduate Student Researcher — Fall 2024 – Present
Programming Systems Group, UC San Diego — *La Jolla, CA*

- Advised by Nadia Polikarpova and Sorin Lerner
- Designed, developed, and evaluated a tool to help programmers explore variation in AI code completions

Visiting Researcher — Summer 2024
CISPA Helmholtz Center for Information Security — *Saarbrücken, Germany*

- Accompanied Prof. Mark Santolucito as a visiting researcher in the Reactive Systems Group
- Developed a framework for synthesizing neurosymbolic AI agents via temporal logic specifications

Research Assistant — Winter 2022 – Spring 2024
Programming Languages Lab, Barnard College — *New York, NY*

- Conducted novel research towards reactive program synthesis as a programming paradigm using JavaScript, TypeScript, and Haskell
- Investigated ideal testbeds for learning specification engineering with temporal logics by integrating reactive synthesis with musical composition, dynamic animation, and game development domains

SWE Research Assistant — Summer 2023
Sunshine Lab, REUSE, Carnegie Mellon University — *Pittsburgh, PA*

- Contributed to Rose, an eDSL for efficient, extensible, web-based automatic differentiation written in Rust and TypeScript, to improve the optimization engine of Penrose, an open-source text-to-visual diagramming system
- Learned state-of-the-art autodiff languages like Google's JAX and Dex and corresponded with Google researchers to contextualize Rose's contributions within existing literature; advised by Sam Estep and Joshua Sunshine

Honors & Awards

Best Paper, IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC) — 2025
Best Short Paper, IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC) — 2025
Graduate Research Fellowship, National Science Foundation (NSF GRFP) — 2024
3rd Place, Student Research Competition, ACM SIGPLAN SPLASH — 2023
PLMW Scholarship, Symposium on Principles of Programming Languages (POPL) — 2023

Publications & Preprints

Decision-Oriented Programming with Aporia — under submission
Saketh Ram Kasibatla*, Raven Rothkopf*, Hila Peleg, Benjamin C. Pierce, Sorin Lerner, Harrison Goldstein, Nadia Polikarpova | arXiv

HiLDe: Intentional Code Generation via Human-in-the-Loop Decoding — VL/HCC 2025
Emmanuel Anaya González*, Raven Rothkopf*, Sorin Lerner, Nadia Polikarpova — Best Paper | DOI

The Command Line GUIde: Graphical Interfaces from Man Pages via AI — VL/HCC 2025
Saketh Ram Kasibatla*, Kiran Medleri Hiremath*, Raven Rothkopf, Sorin Lerner, Haijun Xia, Brian Hempel — Best Short Paper | DOI

Connecting the Dots: Evaluating Abstract Reasoning Capabilities of LLMs Using the NYT Connections Word Game — EMNLP 2024
Prisha Samdarshi, Mariam Mustafa, Anushka Kulkarni, Raven Rothkopf, Tuhin Chakrabarty, Smaranda Muresan | DOI

Rose: Composable Autodiff for the Interactive Web ECOOP 2024
Sam Estep, Wode Ni, Raven Rothkopf, Josh Sunshine | DOI

Towards the Usability of Reactive Synthesis: Mapping TSL Misconceptions to Mitigations PLATEAU 2024
Leyi Cui*, Raven Rothkopf*, Mark Santolucito | DOI

Enforcing Temporal Constraints on Generative Agents with Reactive Synthesis PLATEAU 2024
Raven Rothkopf, Hannah Tongxin Zeng, Mark Santolucito — *presented* | arXiv

Rose: Extensible Autodiff on the Web SPLASH SRC 2023
Raven Rothkopf — 3rd Place Award | DOI

Towards the Usability of Reactive Synthesis: Building Blocks of Temporal Logic PLATEAU 2023
Raven Rothkopf, Leyi Cui, Arya Sinha, Mark Santolucito | DOI

* Equal contribution

Leadership & Service

Artifact Evaluation Committee, TACAS 2025

Student Volunteer, PLDI 2024, OOPSLA 2024, PLATEAU 2025 2024 – 2025

Student Liaison Fall 2022 – Spring 2024
Computer Science Department, Barnard College *New York, NY*

- Led weekly CS Social Hour to uplift women and gender minorities in CS, engaging 50+ attendees; collaborated with the department head to organize program planning meetings for a record 30+ attendees
- Deployed and managed a mentorship program to demystify challenges in CS such as getting into research, exploring fields through introductory workshops, and applying to internships

Skills

Programming: TypeScript, JavaScript, Python, React, Lean, Haskell, LaTeX, Accessible Web Development (HTML, CSS), Agentic Programming Workflows
Tools/Frameworks: Git & GitHub, Figma, vLLM, Z3, Unity, Docker, JAX, Microsoft Office
Research Methods: Contextual Inquiry, Grounded Theory, Interviews, Surveys, Software Usability Testing, Thematic Analysis, Statistical Analysis